BOOKLEAF PUBLISHING

# Technical Assignment

## Full-Stack Developer Position

**Assignment 2 of 2**

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|---|---|
| **Document:** | Author Royalty Dashboard & Automated Notification System |
| **Platform:** | Bubble.io + n8n + AI API |
| **Duration:** | 7 days from date of receipt |
| **Deliverables:** | Bubble app, n8n workflow export, documentation |
| **Confidentiality:** | Do not share publicly |

*This assignment uses a different technology stack than Assignment 1. That is intentional. Please read the full document before beginning.*

## CONTEXT

Congratulations on clearing the first technical round. This second assignment evaluates a different dimension of the role.

BookLeaf's entire operations platform — author dashboards, book management, order processing, production tracking — currently runs on Bubble.io. While we are planning a long-term migration to a fully code-based platform, Bubble will remain our primary operational tool for the foreseeable future. Whoever joins our team will need to build on, maintain, and improve this Bubble-based system day to day.

Alongside Bubble, we rely heavily on n8n for workflow automation — connecting our systems, triggering notifications, syncing data, and integrating AI into our operational processes.

This assignment tests your ability to work with both.

> ⚠️ If you've never used Bubble before, don't worry — that's partly the point. We've given you 7 days (instead of 5) specifically to account for the learning curve. We're evaluating how quickly you can pick up a new platform, how well you understand its logic, and how effectively you build with it. We are not expecting pixel-perfect design.

## THE ASSIGNMENT

Build an **Author Royalty Dashboard & Automated Notification System** using Bubble.io as your application platform and n8n as your automation backend. The system should let authors view their royalty data and let admins trigger AI-generated royalty summary emails to authors — all wired together through automated workflows.

## DETAILED REQUIREMENTS

### A. Bubble Application

Build the entire application in Bubble. This includes the database, UI, user authentication, and internal workflows.

#### A1. Database Design

Set up your Bubble database to store the provided dataset. We're looking at how you structure the data types, define field types, and set up relationships. Think about it the way you'd think about a relational database — authors have books, books have royalty records, etc.

- Import or manually enter the same bookleaf_sample_data.json (provided with Assignment 1) into your Bubble database. You do not need a new dataset — use the one you already have.
- Your database should cleanly separate Authors, Books, and any other entities you think are necessary.

- All royalty figures, sales data, and statuses from the dataset should be queryable and correctly linked.

**A2. Author Portal**

A simple, functional dashboard for authors.

- **Login:** Email/password authentication. Each author sees only their own data.
- **My Books:** A page listing the author's books with key details — title, ISBN, genre, status, publication date, MRP, and platform availability.
- **Royalty Overview:** For each book, show: total copies sold, total royalty earned, royalty already paid, royalty pending, and the last payout date. Also show a consolidated summary across all their books (total earned, total paid, total pending).
- **Royalty Status Indicator:** A clear visual indicator for each book — something like a green/yellow/red badge: Green if royalties are fully paid up, Yellow if there's a pending amount but it's within the normal payout cycle, Red if royalties are overdue (e.g., pending amount exists and last payout was more than 90 days ago, or no payout has ever been made for a published book older than 90 days).

**A3. Admin Portal**

An internal view for the BookLeaf operations team.

- **Author Overview:** A list of all authors with summary stats — total books, total royalty earned, total pending payouts. Filterable by city, by payout status (fully paid / pending / overdue), and searchable by author name.
- **Author Detail View:** Click on any author to see their full book list, detailed royalty figures, and payout history — the same data the author sees, but the admin can see it for everyone.
- **Trigger Royalty Summary:** A button on each author's detail page: "Generate & Send Royalty Summary." When clicked, this triggers an n8n webhook that kicks off the automation described in Part B below.
- **Bulk Trigger:** A button on the main admin page to trigger royalty summaries for all authors with pending royalties in one go. This should call the same n8n webhook, once per eligible author.
- **Notification Log:** A simple log showing which authors have been sent royalty summaries, when, and the status (Success / Failed). This should update after n8n completes its workflow.

**A4. Privacy Rules**

Set up Bubble's privacy rules so that author-type users can only access their own data. Admin-type users can access all data. We will test this by logging in as different authors and verifying data isolation.

### B. n8n Automation Workflow

Build an n8n workflow that gets triggered from the Bubble app and handles the AI-powered royalty summary generation. Here's the flow we expect:

| Step | Description |
|---|---|
| 1. Webhook | n8n receives a webhook call from Bubble containing the author's ID (or relevant identifier). |
| 2. Fetch Data | n8n pulls the author's complete data — either by querying the Bubble Data API or by receiving the full payload in the webhook. Either approach is acceptable; justify your choice. |
| 3. AI Processing | n8n sends the author's royalty data to an AI API (OpenAI, Anthropic, or your choice) with a well-crafted prompt to generate a professional, human-readable royalty summary email. |
| 4. Output | The generated email content is either: (a) sent as an actual email via an SMTP/email node (bonus), or (b) sent back to Bubble via an API call and logged in the Notification Log (required). Either way, the output must be visible and verifiable. |
| 5. Status Update | n8n sends a callback to Bubble updating the notification log with Success or Failed status, along with a timestamp. |

### What the AI-Generated Royalty Summary Should Contain

The AI should generate an email that an author would actually find useful. It should include:

- A personalised greeting using the author's name.
- A summary of each of their published books with sales and royalty figures.
- A clear breakdown: total earned, total paid, total pending.
- If there's a pending amount, an explanation of when they can expect the next payout (based on the quarterly cycle and 45-day window from the Knowledge Base in Assignment 1).
- If all royalties are paid up, a positive acknowledgement.
- If any books are still in production (not yet published), a brief status update.
- A warm, professional closing that matches BookLeaf's communication tone.

> Refer to the BookLeaf Knowledge Base and Communication Tone Guidelines provided in Assignment 1. The same tone and policy accuracy expectations apply here.

### C. Integration & Error Handling

This is where we see how you think about systems working together in the real world.

- **Webhook Security:** The n8n webhook should not be openly callable by anyone. Implement some form of authentication — an API key header, a shared secret, or Bubble passing a token that n8n validates.

- **Error Handling in n8n:** What happens if the AI API call fails? What if Bubble's Data API is unreachable? Your workflow should handle failures gracefully — log the failure, update the notification status to "Failed," and not break silently.
- **Error Handling in Bubble:** If the n8n webhook doesn't respond or returns an error, the Bubble UI should reflect this. Don't leave the admin staring at a loading spinner forever.
- **Idempotency:** If an admin accidentally clicks "Generate & Send" twice for the same author in quick succession, the system should handle this sensibly — either prevent duplicate triggers or de-duplicate on the n8n side.

## SAMPLE DATASET

Use the **same bookleaf_sample_data.json** provided with Assignment 1. Import this data into your Bubble database. Your app should work with these exact records — we will test using this data.

> Quick reference: The dataset contains 10 authors and 18 books, including:
> - Authors with multiple books and active royalty payouts
> - Authors with pending royalties that have never been paid out
> - Books still in production (no sales/royalty data yet)
> - Books across different print partners (In-House, Repro India, Epitome Books)
>
> Your system should handle all these variations correctly.

## TECHNOLOGY NOTES

### Bubble.io

- Bubble's free plan is sufficient for this assignment. You do not need to be on a paid plan.
- If you're new to Bubble, their official manual (manual.bubble.io) and YouTube tutorials are the best starting points. Most candidates find the learning curve manageable within 2–3 days.
- Focus on database design and workflows first, UI polish second. We'd rather see a well-structured app with a basic UI than a pretty app with a messy database.

### n8n

- You can use n8n Cloud (free trial available) or self-host it locally. Either is fine.
- If you self-host, make sure the webhook URL is accessible from the internet (use a tool like ngrok if needed) so that Bubble can call it.
- Export your workflow as a JSON file — this is one of your deliverables.

### AI API

- Same guidance as Assignment 1: use OpenAI, Anthropic, or any LLM API you prefer.
- Use your own API key. Budget-friendly options like gpt-4o-mini or Claude Haiku are perfectly acceptable.

## WHAT TO SUBMIT

1. **Bubble Application:** Share editor access to your Bubble app by adding [we will provide the email] as a collaborator. Also provide the live app URL with test login credentials (one author account, one admin account).
2. **n8n Workflow Export:** Export your n8n workflow as a JSON file and include it in your submission. If using n8n Cloud, also provide view access to your workspace.

3. **Documentation (README):** A document or PDF covering: your Bubble database schema and why you structured it that way, how the Bubble ↔ n8n integration works, your prompt strategy for the AI-generated royalty summaries, error handling approach, and any challenges you faced (especially if Bubble was new to you — we want to understand your learning process).
4. **Screen Recording (3–5 minutes):** A short Loom or screen recording walking through the app: show the author portal, the admin portal, trigger a royalty summary, and show the n8n workflow executing. This helps us see the full flow even if there are deployment hiccups.

> 💡 The screen recording is important. Bubble apps on free plans can be slow, and n8n Cloud trials may expire. The recording ensures we can evaluate your work even if the live environment has issues during our review.

## EVALUATION CRITERIA

| Criteria | Weight | What We're Looking For |
|---|---|---|
| Bubble Proficiency | 25% | Database design, privacy rules, workflows, API connector usage, repeating groups, conditionals. If you're new to Bubble, we're evaluating how well you learned it — not prior experience. |
| n8n Workflow Design | 25% | Webhook setup, data flow, error handling, node organisation, use of n8n features (error branches, IF nodes, function nodes). Is the workflow clean, logical, and robust? |
| AI Integration & Output Quality | 20% | Prompt design, quality of generated royalty summaries, use of Knowledge Base context, handling of edge cases (authors with no payouts, books in production). |
| Integration Quality | 15% | How well Bubble and n8n talk to each other. Webhook security, callback handling, status updates, error propagation between the two systems. |
| Documentation & Learning | 15% | README quality, schema explanation, learning process documentation. If Bubble was new to you, we want to see how you approached the learning curve. |

## TIMELINE & PROCESS

You have **7 days** from the date you receive this assignment. The extra time (compared to Assignment 1) is to account for the Bubble learning curve. If you already know Bubble, you'll likely finish faster — use the extra time to polish your n8n workflow and documentation.

After submission, if your work clears our evaluation, the final step is a conversation with the founders. This will cover both your assignments, your technical thinking, and a broader discussion about how you'd approach the role at BookLeaf.

## IMPORTANT NOTES

- The same confidentiality rules from Assignment 1 apply. Do not share this assignment publicly.
- You are free to use AI tools during development. But you must be able to explain your Bubble workflows and n8n setup in detail during the founder conversation.
- If you get stuck on Bubble specifically, that's fine — document what you tried, what didn't work, and how you worked around it. We value the learning process, not just the output.
- If anything is unclear, email us. Same as before — asking questions is a positive signal.

**You're in the final stretch. Show us how you learn, build, and connect systems.**

Team BookLeaf